Product Requirements Document
AI Customer Support Assistant 1793813526269

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1. Document Information

Document Title: Product Requirements Document for AI Customer Support Assistant 1793813526269

Document ID: PRD-6AEB6C18

Version: 1.0 (Draft)

Target Domain: Customer Support

Product Category: Customer Support

Status: Under Active Review

2. Executive Summary

AI Customer Support Assistant 1793813526269 addresses key workflow challenges within Customer Support. This document provides the authoritative engineering specification to implement the complete solution, ensuring local data sovereignty, deterministic execution, and rapid user delivery.

3. Product Overview

Vision
To deliver an autonomous, highly reliable system solving core operational pain points in Customer Support.

Key Capabilities
- Knowledge-base article retrieval
- Conversation history management
- Uncertain answer identification
- Escalation to human support agents

4. Problem Statement

Current practitioners in Customer Support experience friction due to manual, fragmented, or error-prone processes.

Identified Pain Points:
- Lack of automated structure and standardization.
- Missing critical requirements and edge-case handling.
- Heavy reliance on fragmented or proprietary external toolchains.

5. Product Vision and Objectives

Core Objectives
1. Answer customer questions
2. Resolve issues efficiently

6. Target Users and Stakeholders

Primary Stakeholders:
- **Customer:** Primary user interacting with system capabilities.
- **Support Agent:** Primary user interacting with system capabilities.

7. User Personas

Persona 1: Primary Practitioner (Customer)
- **Goals:** Rapidly execute domain workflows with minimal manual boilerplate.
- **Frustrations:** Incomplete documentation, ambiguous requirements, and manual rework.
- **Technical Proficiency:** Intermediate to Advanced.

8. Existing Workflow and Pain Points

1. **Manual ideation & drafting:** Practitioners write unstructured notes.
2. **Information gaps:** Crucial architectural constraints and non-functional requirements are forgotten.
3. **Inconsistent documentation:** Hand-offs to engineering teams fail due to missing acceptance criteria.

9. Proposed Solution

AI Customer Support Assistant 1793813526269 introduces an end-to-end automated framework that bridges raw concepts into verified, production-grade requirements. It features structured pipeline processing, interactive clarification, and deterministic quality validation.

10. Product Scope

In-Scope Capabilities for MVP:
- Complete workflow automation from user input to structured output.
- Local-first execution without external SaaS dependencies.
- Interactive clarification and assumption tracking.
- Deterministic quality validation and scoring.
- Multi-format export (Markdown, PDF, DOCX).

11. Out of Scope

Non-Goals (Out of Scope for Initial Release):
- Automatic code generation or deployment of arbitrary target applications.
- Third-party cloud SaaS synchronization.
- Automated financial market viability forecasting.

12. Functional Requirements

ID	Title	Description	Priority	Traceable AC
FR-001	Natural Language Input Processing	The system shall ingest and parse unstructured user product descriptions.	Must Have	AC-001
FR-002	Automated PRD Synthesis	The system shall synthesize functional and non-functional requirements into a canonical 27-section PRD.	Must Have	AC-002

13. Non-Functional Requirements

ID	Title	Description	Priority
NFR-001	Local Host Confidentiality	All data processing and persistence shall occur strictly on local host storage.	Must Have

14. User Stories

Story ID	As a...	I want to...	So that...	Linked FR
US-001	Customer	natural language input processing	The system shall ingest and parse unstructured user product descriptions.	FR-001
US-002	Support Agent	automated prd synthesis	The system shall synthesize functional and non-functional requirements into a canonical 27-section PRD.	FR-002

15. User Journeys

```mermaid
sequenceDiagram
actor User
participant Framework as PRD Framework
participant Pipeline as AI Processing Pipeline
participant Validator as Quality Engine

User->>Framework: Submit concept / assignment
Framework->>Pipeline: Analyze input & identify gaps
Pipeline-->>User: Clarification questions & assumptions
User->>Pipeline: Confirm answers / accept defaults
Pipeline->>Framework: Synthesize 27 canonical sections
Framework->>Validator: Run deterministic checks
Validator-->>User: Validation report & quality score
User->>Framework: Edit/Approve & Export (MD/PDF/DOCX)
```

16. Business Rules

ID	Rule Title	Operational Statement
BR-001	Immutable Version Tracking	Approved PRD versions shall remain immutable to preserve auditability.
BR-002	Customer Data Privacy	All customer data must be stored securely and in compliance with GDPR regulations. Access to customer data must be restricted to authorized personnel only.
BR-003	Issue Resolution Process	All customer issues must be resolved within 24 hours of initial report. The resolution process must be documented and stored in the customer support database.
BR-004	Compliance with Industry Regulations	All customer support interactions must be recorded and stored in accordance with industry regulations (e.g. HIPAA, PCI-DSS).
BR-005	Data Validation	All customer data must be validated against a set of predefined rules to ensure accuracy and completeness. For example, email addresses must be in a valid format.
BR-006	Operational Requirements	The customer support system must be available 24/7 and must have a minimum of 2 support agents online during peak hours. The system must also have a built-in escalation process for complex issues.

17. System Workflows

1. **Input Ingestion:** Receive text and attachments; persist raw input record.
2. **Domain Classification:** Parse assignment context or problem domain.
3. **Gap Resolution:** Clarify critical missing attributes or default to documented assumptions.
4. **Requirement Synthesis:** Formulate FRs, NFRs, BRs, and testable ACs.
5. **Audit & Export:** Validate schema, verify ID uniqueness, render document files.

18. Data Requirements

- **Persistence Model:** Document-oriented collections with immutable version archives.
- **Data Integrity:** Strict foreign key references between requirements, acceptance criteria, and version snapshots.
- **Storage Isolation:** File attachments and exported documents reside in local system storage directories.

19. Integration Requirements

- **Local AI Gateway:** REST communication with local Ollama daemon (`http://localhost:11434`).
- **Offline Fallback Engine:** In-process deterministic execution adapter when external inference engines are unavailable.
- **File Formats:** Direct binary writers for Markdown, DOCX (Office OpenXML), and PDF.

20. Security and Privacy Requirements

- **Data Sovereignty:** All telemetry, database records, and prompts remain strictly within the local host boundary.
- **Authentication:** JWT tokens with cryptographic hashing (bcrypt) for local access protection.
- **Input Sanitation:** File uploads are validated and isolated from execution paths.

21. Acceptance Criteria

AC ID	Associated Requirement	Acceptance Criteria (Given-When-Then / Rule)
AC-001	FR-001	Given a submitted prompt, when saved, the system creates an identifiable submission record.
AC-002	FR-002	Given valid analysis inputs, when PRD generation runs, all canonical sections are generated.
AC-003	NFR-001	Given execution in an offline environment, all operations complete without external network requests.

22. Dependencies and Constraints

- **Runtime Dependencies:** Python 3.10+, Node.js 18+, MongoDB Community / in-memory instance, Ollama.
- **Hardware Constraints:** Minimum 8GB RAM recommended for local 3B/7B model execution.
- **Zero-Cloud Constraint:** The application must remain fully functional without outbound internet access.

23. Risks and Mitigation

Risk	Category	Impact	Mitigation Strategy
Local inference timeout	Performance	High	Asynchronous job execution and fallback offline generation adapter.
Hallucinated business facts	Quality	High	Strict boundary separation between confirmed facts and labeled assumptions.
Accidental version overwrite	Data Integrity	Critical	Immutable version snapshots; all edits produce incremental versions.
Incomplete requirement coverage	Compliance	Medium	Deterministic validation engine flagging missing sections or ACs.

24. Assumptions and Open Questions

Documented Assumptions
- [Assumed] AI can understand natural language
- [Assumed] Knowledge-base articles are up-to-date and accurate

Open Questions for Future Iterations
- Will real-time multi-peer collaboration be required in future versions?
- Is automated sync to issue trackers (Jira/GitHub) desirable in subsequent milestones?

25. Success Metrics

- **Document Generation Completeness:** 100% of canonical sections populated or documented as N/A.
- **Traceability Score:** 100% of Functional Requirements linked to at least one testable Acceptance Criterion.
- **Validation Pass Rate:** Zero critical validation errors prior to version approval.
- **Offline Operation:** Zero external network requests initiated during standard execution.

26. Release Considerations

- **Deployment Formats:** Single-command Docker Compose stack and native local environment scripts.
- **Data Migration:** Automated schema migration hooks on service startup.
- **Backward Compatibility:** Versioned API endpoints under `/api/v1`.

27. Appendix and Glossary

- **PRD:** Product Requirements Document.
- **FR:** Functional Requirement.
- **NFR:** Non-Functional Requirement.
- **BR:** Business Rule.
- **AC:** Acceptance Criteria.
- **Local-First:** Software architecture prioritizing local execution and data ownership.